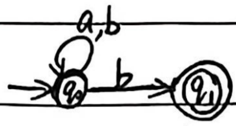
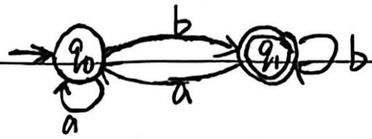
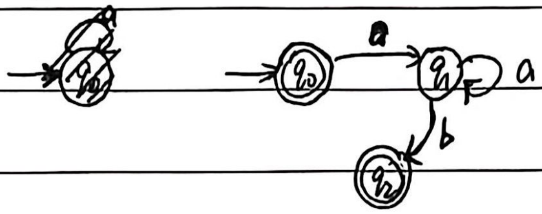
编译原理 Assignment 2.

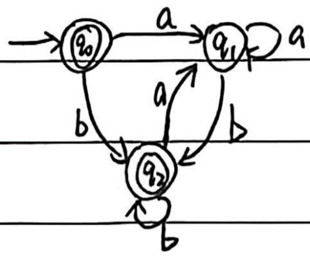
Exercise 1

1. $L((a|b)^*b)$: NFA => 

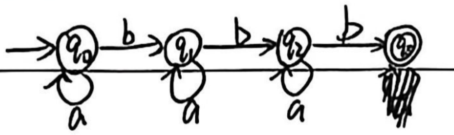
DFA => 

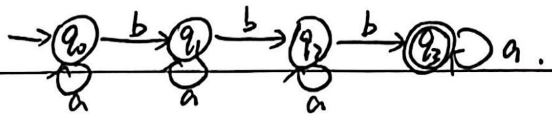
2. $L(((\varepsilon|a)^*b)^*)$ NFA => 

$\Leftrightarrow L((a^*b)^*)$

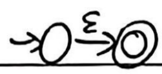
DFA => 

3. $L(a^*ba^*ba^*ba^*)$

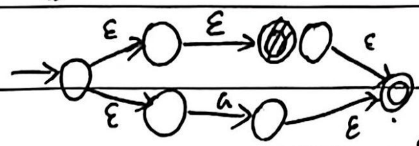
NFA => 

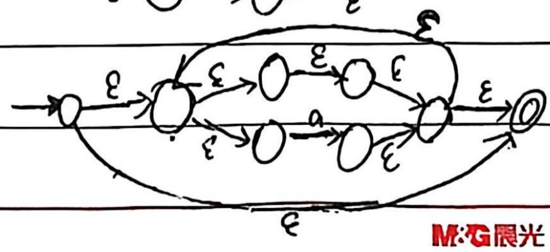
DFA => 

Exercise 2.

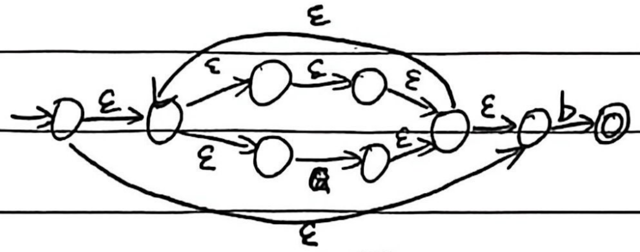
1. $((\varepsilon|a)^*b)^*$: $N(a)$:  $N(\varepsilon)$: 

$N(b)$: 

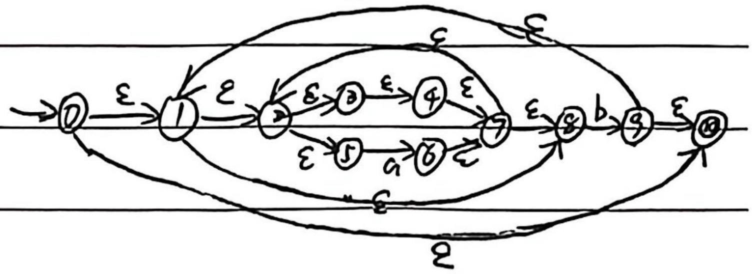
$N((\varepsilon|a))$: 

$N(((\varepsilon|a)^*))$: 

N(a((ε|a)*b)) :



N(((ε|a)*b)*) :

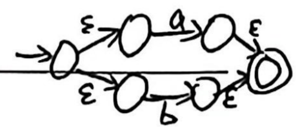


2. (a|b)*a(a|b)(a|b).

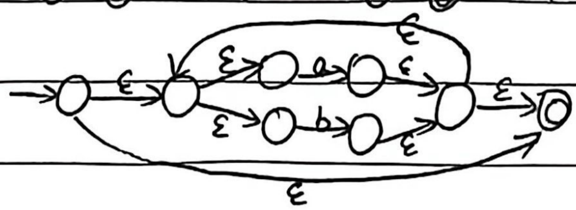
N(a) :

N(b) :

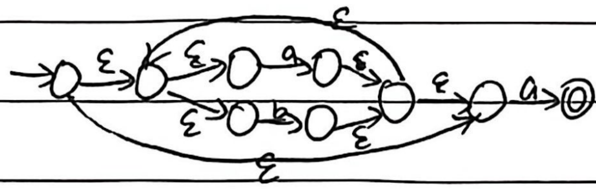
N(a|b) :



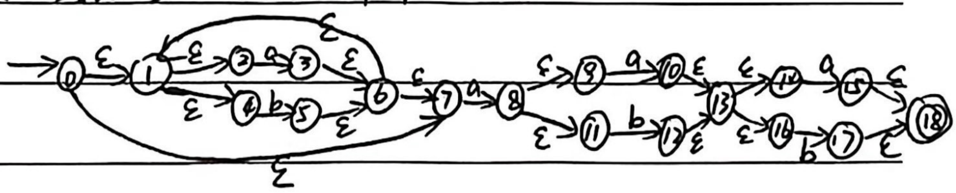
⇒ N((a|b)*) :



⇒ N((a|b)*a) :



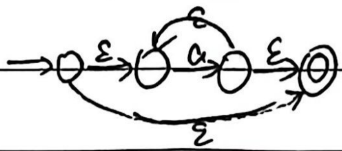
⇒ N((a|b)*a (a|b)(a|b)) (重复操作,连接).



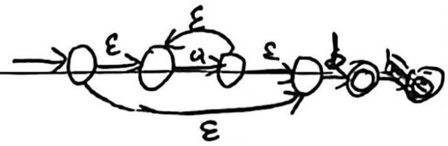
3. a*ba*ba*ba*

N(a) : →○ —a→ ◎ N(b) : →○ —b→ ◎

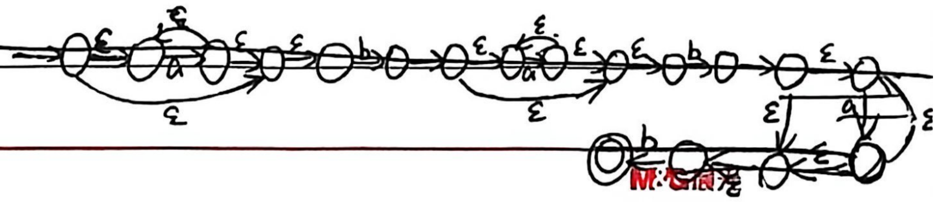
N(a*) :



N(a*b) :

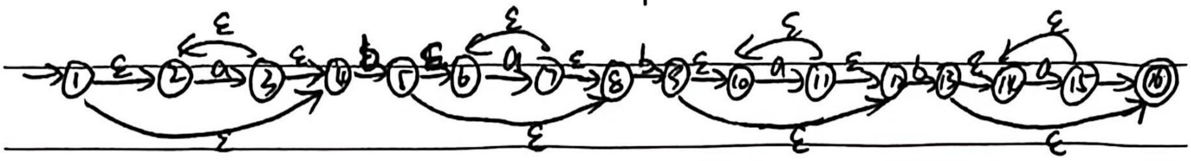


N(a*ba*ba*b) :



~~该连接操作~~

N(a*ba*ba*ba*)： 重复连接操作.



Exercise 3.

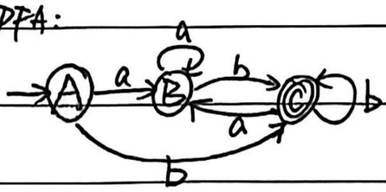
1. A: ε(0) = {0,1,2,3,4,5,7,8,10}.

B: ~~D~~ Dtran[A,a] = ε({6}) = ~~{6,7,8}~~ {2,3,4,5,6,7,8}.

C: Dtran[A,b] = ε({9}) = {1,2,3,4,5,7,8,9,10}

DFA state	a	b
A	B	C
B	B	C
C	B	C

DFA:



2. A: ε(0) = {0,1,2,4,7}

B: Dtran[A,a] = ε({3,8}) = {1,2,3,4,6,7,8,9,11}.

C: Dtran[A,b] = ε({5}) = {1,2,4,5,6,7}

D: Dtran[B,a] = ~~ε({10}) = {10,13,14,16}~~ ε({3,8,10}) = {1,2,3,4,6,7,8,9,10,11,13,14,16}

E: Dtran[B,b] = ε({5,12}) = {1,2,4,5,6,7,12,13,14,16}.

~~E~~ Dtran[~~A~~C,a] = ε({3,8}) ⇒ B.

Dtran[C,b] = ε({5}) ⇒ C.

F: Dtran[D,a] = ε({3,8,10,15}) = {1,2,3,4,6,7,8,9,10,11,13,14,15,16,18}

G: Dtran[D,b] = ε({5,12,17}) = {1,2,4,5,6,7,13,14,16,17,18}.

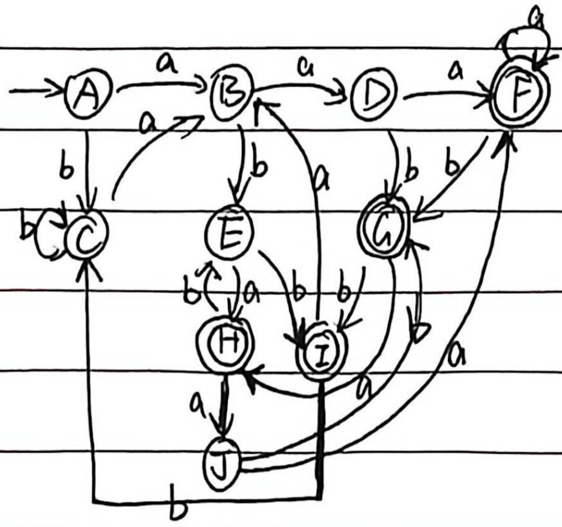
H: Dtran[E,a] = ε({3,8,15}) = {1,2,3,4,6,7,8,9,11,15,18}

I: Dtran[E,b] = ε({5,17}) = {1,2,4,5,6,7,17,18}.

J: Dtran[H,a] = ε({3,8,10}) = {1,2,3,4,6,7,8,9,11,13,14,16}.

DFA State	a	b
A	B	C
B	D	E
C	B	C
D	F	G
E	H	I
F	F	G
G	H	I
H	J	E
I	B	C
J	F	G

DFA:



3. A: $\varepsilon(\{1\}) = \{1,2,4\}$.

B: $Dtran[A,a] = \varepsilon(\{3\}) = \{2,3,4\}$. C: $Dtran[A,b] = \varepsilon(\{5\}) = \{5,6,8\}$.

$Dtran[B,a] = \varepsilon(\{3\}) \Rightarrow B$ $\quad$ $Dtran[B,b] = \varepsilon(\{5\}) \Rightarrow C$.

D: $Dtran[C,a] = \varepsilon(\{7\}) = \{6,7,8\}$. E: $Dtran[C,b] = \varepsilon(\{9\}) = \{9,10,12\}$.

$Dtran[D,a] = \varepsilon(\{7\}) \Rightarrow D$ $\quad$ $Dtran[D,b] = \varepsilon(\{9\}) \Rightarrow E$.

F: $Dtran[E,a] = \varepsilon(\{11\}) = \{10,11,12\}$, G: $Dtran[E,b] = \varepsilon(\{13\}) = \{13,14,16$

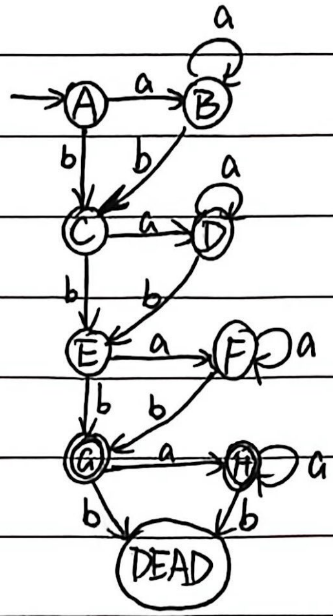
$Dtran[F,a] = \varepsilon(\{11\}) \Rightarrow F$ $\quad$ $Dtran[F,b] = \varepsilon(\{13\}) \Rightarrow G$.

H: $Dtran[G,a] = \varepsilon(\{15\}) = \{14,15,16\}$. $Dtran[G,b] = \varepsilon(\phi) = \phi$ $\boxed{\text{DEAD}}$.

$Dtran[H,a] = \varepsilon(\{15\}) \Rightarrow H$ $\quad$ $Dtran[H,b] \Rightarrow DEAD$.

DFA State	a	b
A	B	C
B	B	C
C	D	E
D	D	E
E	F	G
F	F	G
G	H	DEAD
H	H	DEAD

DFA:



Optional Exercise 1.

1. 首先分组为 $G_1=\{A,B\}$, $G_2=\{C\}$.

对 G_1:

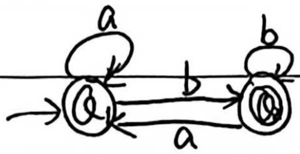
DFA State	a	b
A	$B(G_1)$	$C(G_2)$
B	$B(G_1)$	$C(G_2)$

不可划分. 最终: $\{\{A,B\},\{C\}\}$.

$\Downarrow$

$\{Q_1, Q_2\}$.

DFA:



2. 首先分为 $G_1=\{A,B,C,D,E,J\}$ $G_2=\{F,G,H,I\}$.

对 G_2:

	a	b
F	$F(G_2)$	$G(G_2)$
G	$H(G_2)$	$I(G_2)$
H	$J(G_1)$	$E(G_1)$
I	$B(G_1)$	$C(G_1)$

划分为 $G_{21}=\{F,G\}$ $G_{22}=\{H,I\}$.

对G_1：

		a	b
	A	B(G_1)	C(G_1)
	B	D(G_1)	E(G_1)
	C	B(G_1)	C(G_1)
	D	F(G_2)	G(G_2)
	E	H(G_2)	I(G_2)
	J	F(G_2)	G(G_2)

划分为 $G_{11}=\{A,B,C\}$

$G_{12}=\{D,E,J\}$.

此时 $\Pi=\{\{A,B,C\},\{D,E,J\},\{F,G\},\{H,I\}\}$.

对G_{22}：

	a	b
H	J(G_{12})	E(G_{12})
I	B(G_{11})	C(G_{11})

划分为$\{H\},\{I\}$.

对G_{21}：

	a	b
F	F(G_{11})	G(G_{11})
G	H(G_{22})	I(G_{22})

划分为$\{F\},\{G\}$.

对G_{12}：

	a	b
D	F	G
E	H	I
J	F	G

划分为$\{D,J\},\{E\}$.

对G_{11}：

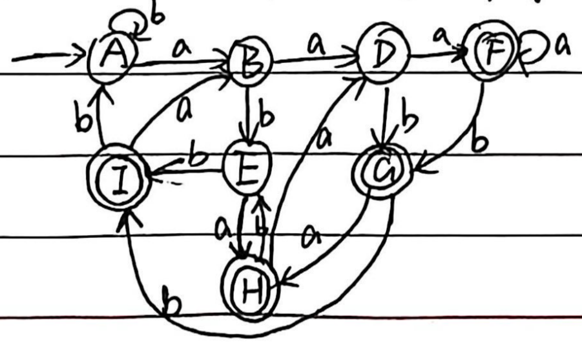
	a	b
A	B	C
B	D	E
C	B	C

划分为$\{A,C\},\{B\}$.

此时 $\Pi=\{\{A,C\},\{B\},\{D,J\},\{E\},\{H\},\{I\},\{F\},\{G\}\}$.

对$\{A,C\}$不可划分. 对$\{D,J\}$不可划分.

则最终DFA：



3. 首次：$G_1 = \{A,B,C,D,E,F\}$ $G_2 = \{G,H\}$

对G_2

	a	b
G	H	DEAD
H	H	DEAD

不划分。

对G_1

	a	b
A	B	C
B	B	C
C	D	E
D	D	E
E	F	G
F	F	G

划分为$\{A,B,C,D\}$，$\{E,F\}$.

此时 $\pi = \{\{A,B,C,D\},\{E,F\},\{G,H\}\}$.

显然 $\{E,F\}$，$\{G,H\}$不可划分.

对$\{A,B,C,D\}$：

	a	b
A	B	C
B	B	C
C	D	E
D	D	E

划分为$\{A,B\}$，$\{C,D\}$.

最终：$\{\{A,B\},\{C,D\},\{E,F\},\{G,H\}\}$.

DFA：